TECHNICAL EVIDENCE MEMO

CHMT-Q Tiny

Retrospective BTC-USDT Forecast Evaluation

Better proper score. No trading conclusion.

5.76%

LOWER FORWARD PINBALL

0.242370 vs 0.257193

282,838

TRAINABLE PARAMETERS

86.46% below reference size

80.19%

Q10-Q90 COVERAGE

Nominal central-band target: 80%

DECISION: CONTINUE RESEARCH. DO NOT ALLOCATE CAPITAL OR CLAIM PROFITABILITY.

Dhruv Sharma | OKX BTC-USDT spot | Previously studied 21-24 June 2026 partition

RESEARCH-ONLY

What the result establishes - and what it does not

AUDITED HEADLINE

On 5,724 overlapping development examples, seed-17 CHMT-Q Tiny produced weighted normalized pinball loss of 0.242370 versus 0.257193 for the registered train-only empirical baseline. The descriptive reduction was 5.7633%, and every one of the 25 horizon-target cells was lower.

Against the best-performing added simple context baseline (trailing three complete days), Tiny was 4.76% lower. That comparison is post hoc, not registered.

1

Marginal quantile score

EVIDENCED

Proper score improved across all 25 cells.

2

Conditional signal

WEAK

H5 total-POC correlation and dispersion remain small.

3

Executable economics

NOT MEASURED

No Tiny cost, fill, PnL, Sharpe, or capacity study.

FUND-FACING INTERPRETATION

A compact hierarchical Transformer achieved lower aggregate weighted normalized pinball loss than several simple retrospective baselines on one accessed period. This is not evidence of alpha, calibration under regime change, architectural superiority, or a profitable strategy.

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Chronology, dependence, and governance

TRAIN	SELECT	DEVELOP	RETIRED
01-18 JUN	19-20 JUN	21-24 JUN	25-30 JUN
25,758	2,862	5,724	not scored

UTC sessions. June 25-30 is process-accessed and retired; it is not a sealed outer holdout.

RETROSPECTIVE STATUS

The development partition was already studied. All added metrics and context baselines in this memo are post hoc and hypothesis-generating.

ROW DEPENDENCE

Adjacent rows share 63 of 64 bars and four of five longest-horizon labels. Row-wise p-values and confidence intervals would be anti-conservative.

TARGET CONTRACT

Five footprint descriptors are forecast at five one-minute horizons. They are not returns, trade actions, or a joint predictive distribution.

Artifact governance

168
Arrow shards hash/schema verified

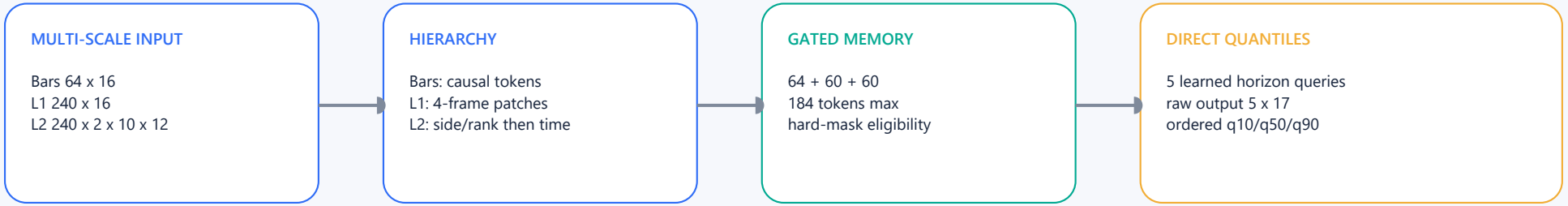
42,930
content-addressed corpus rows

1.725 GB
verified Arrow bytes

0
accepted nonfinite outputs

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Causal hierarchy, hard masks, quality gates, direct quantiles



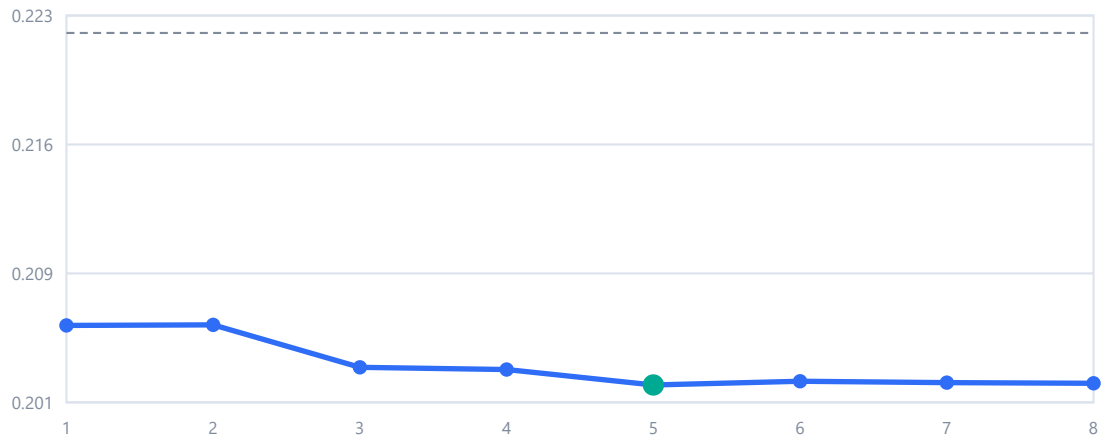
Subsystem	Tiny setting	Reference setting	Interpretation
Shared width	64	128	Capacity halved
L2 level width	16	32	Side/rank hierarchy retained
Encoder depth	1 per branch	3 per branch	Shorter temporal stacks
Decoder depth	2	2	Five direct horizons retained
Parameters	282,838	2,089,554	86.46% fewer
Presence logits	2 inactive slots	2 trained slots	Tiny loss weight fixed to zero

Causal means temporal availability and masking, not causal inference. Hard masks determine eligibility; soft reliability gates cannot restore unavailable evidence.

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Selection curve, objective, and reproducibility ledger

Validation selection curve



Epoch (teal marks selected epoch 5; dashed line is empirical baseline)

Objective and optimizer

Primary	weighted normalized pinball
Auxiliary	median Huber + deep supervision
Presence	weight 0; labels 100% positive
Optimizer	AdamW, matrix decay only
Learning rate	3e-4 cosine to 3e-5
Effective batch	32 x 4 = 128
Gradient clip	global norm 1.0
Selection	epoch 5 of 8; seed 17

25.9 min

TRAIN WALL TIME

Local shared GPU environment

40.65 ms

BATCH-ONE LATENCY

Shared GTX 1650; not an SLA

0.0

REPEAT MAX ABS DIFF

Two identical inference calls

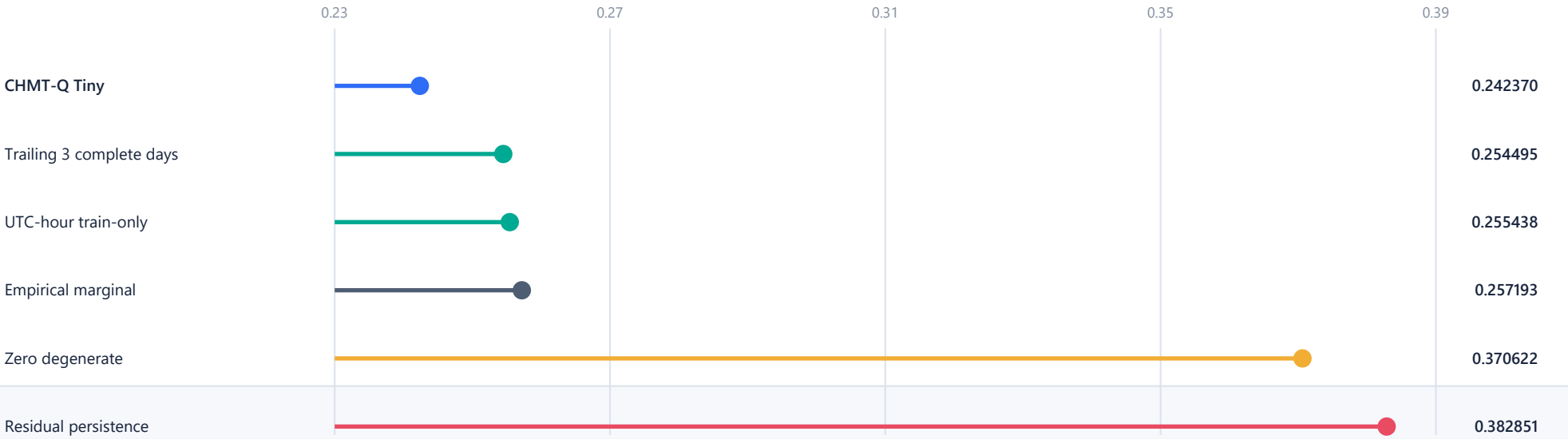
1,616

OPTIMIZER UPDATES

Eight completed epochs

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Registered result plus post hoc context baselines



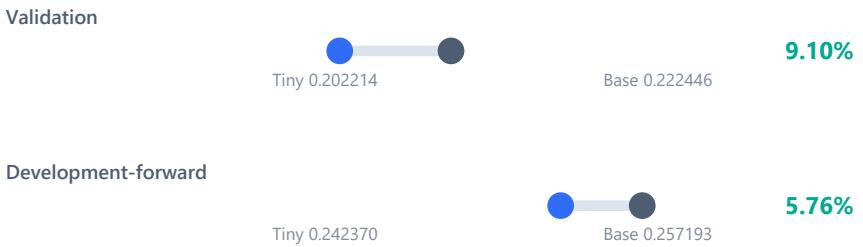
Benchmark	Pinball	Skill vs empirical	Coverage	Status
CHMT-Q Tiny	0.242370	+5.76%	80.19%	selected on validation
Trailing 3 complete days	0.254495	+1.05%	76.39%	post hoc adaptive
UTC-hour train-only	0.255438	+0.68%	83.81%	post hoc context
Empirical marginal	0.257193	+0.00%	83.68%	registered baseline
Zero degenerate	0.370622	-44.10%	1.24%	post hoc sanity check
Residual persistence	0.382851	-48.86%	83.84%	post hoc causal anchor

Score composition: total/buy/sell POC receive 75% of target weight; log-volume and delta receive 12.5% each; horizon weights decay as 1/sqrt(h). Only the empirical marginal comparator was frozen before the original evaluation. The four added baselines are post hoc context, not new evidence.

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Aggregate gains and all 25 horizon-target cells

Tiny versus registered empirical baseline



25 / 25

CELL WINS

Tiny below empirical baseline

0.90%-7.19%

CELL SKILL RANGE

No target/horizon cell reversed

Cell-level pinball skill versus empirical (%)

	Total	Buy	Sell	Log-vol	Delta
H1	7.2	7.1	7.1	3.4	1.7
H2	7.1	7.0	6.8	2.0	1.1
H3	6.5	6.4	6.4	1.5	1.2
H4	6.4	6.2	6.2	1.6	1.2
H5	6.0	5.9	6.0	1.9	0.9

Mean skill decomposition



Gain narrows with horizon: 5.29% at H1 to 4.14% at H5 on an equal-cell average.

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Coverage, sharpness, quantile hits, and output-path checks

80.19%
TINY COVERAGE
Q10-Q90 central band

83.68%
EMPIRICAL COVERAGE
Wider is not automatically better

+0.19 pp
NOMINAL GAP
Descriptive only; target is 80%

0 / 0
CROSSINGS / SATURATION
No quantile crossing; delta tails unsaturated

Tiny q10-q90 coverage by cell

	Total	Buy	Sell	Log-vol	Delta
H1	81%	81%	81%	82%	75%
H2	82%	82%	82%	82%	76%
H3	81%	81%	82%	81%	76%
H4	81%	81%	81%	81%	76%
H5	81%	81%	81%	81%	75%

Target-mean diagnostics

Target	q10 hit	q50 hit	q90 hit	Norm width	Interval score
Total	9.0%	50.7%	90.4%	2.48	3.77
Buy	9.4%	51.7%	90.7%	2.50	3.82
Sell	9.5%	52.1%	90.9%	2.48	3.75
Log-vol	9.2%	54.3%	90.6%	2.60	3.65
Delta	11.5%	48.9%	87.1%	1.52	1.78

Coverage near 80% does not prove calibration. The 25 cells differ materially, and interval width must be reported in target-specific units or normalized scale - never averaged across ticks, log-volume, and fractions.

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Proper-score gains coexist with weak H5 location signal

H5 total-POC	MAE ticks	RMSE	Pearson	Spearman	Dispersion	Sign MCC
Tiny	531.8	769.2	0.046	0.057	0.071	0.042
Empirical	532.9	769.9	n/a	n/a	0.000	n/a
UTC hour	533.6	769.3	0.041	0.030	0.047	0.006
Trailing 3d	533.1	769.8	0.027	0.022	0.015	n/a
Persistence	840.5	1239.0	-0.045	-0.025	1.217	-0.023

0.046

PEARSON

Weak linear association

0.057

SPEARMAN

Weak rank association

0.071

DISPERSION RATIO

Medians strongly compressed

0.042

SIGN MCC

Not an executable direction test

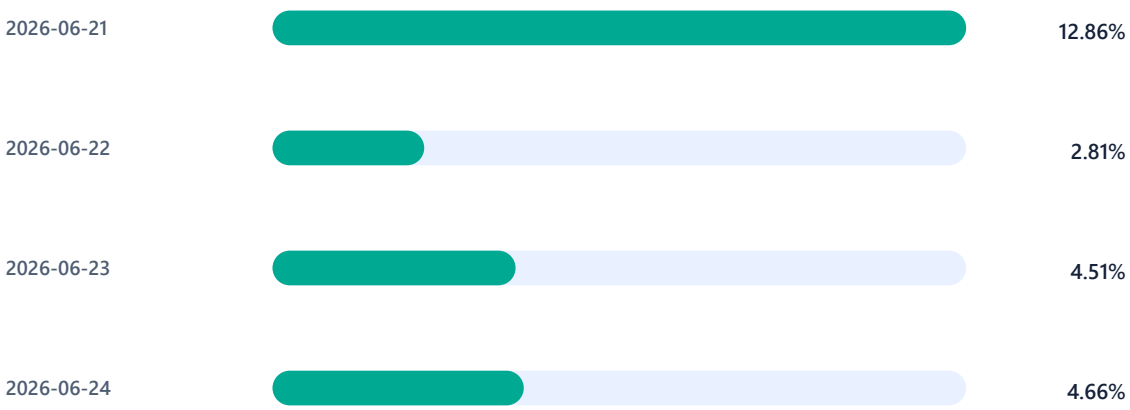
DECISION-UTILITY GAP

Tiny improves the marginal quantile score and H5 total-POC MAE by about 0.20% versus the empirical median, but its H5 medians show weak association with realized variation. Predicted-median standard deviation is only 7.1% of realized standard deviation. Better proper scoring is not equivalent to useful direction, timing, or edge.

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Four forward days and a 64-phase overlap sensitivity

Daily relative improvement versus empirical



All four days favored Tiny, but four days are not an inferential sample.

64-minute phase sensitivity

64 / 64

FAVORABLE PHASES

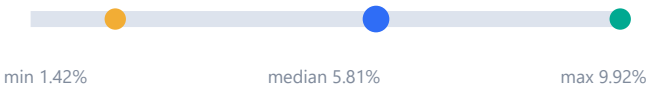
One row every 64 minutes

5.81%

MEDIAN SKILL

Across clock phases

Relative improvement range



WHY THIS IS A SENSITIVITY, NOT A CONFIDENCE INTERVAL

The phase check samples one row every 64 minutes for each possible clock phase, reducing shared 64-minute bar context. Every phase remains favorable, which is useful robustness context. It does not solve market nonstationarity, target overlap, model-selection reuse, or the fact that only four UTC days are observed. No row-wise p-value, Sharpe inference, or bootstrap claim is reported.

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One input/output path - illustrative, not representative

okx-btc-usdt-20260621T000500Z

`bars [64, 16] | L1 [240, 16] | L2 [240, 2, 10, 12] | quality [16]`

4	240	4800	124
valid bars	valid L1 frames	valid L2 cells	eligible memory tokens

Horizon-1 target	q10	q50	q90	Realized	Inside band
Total POC ticks	-116.149	8.017	133.046	0.500	yes
Buy POC ticks	-115.785	15.923	141.257	0.500	yes
Sell POC ticks	-97.812	26.542	160.080	72.500	yes
Log-volume residual	-0.656	-0.098	0.853	-0.099	yes
Delta fraction	-0.866	-0.071	0.799	0.441	yes

Quality-bounded modality gates

0.363	0.224	0.412
Bars	L1	L2

DO NOT READ THIS ROW AS A TRADE

The five outputs are footprint descriptors, not returns or actions. Presence logits were inactive. A single trace demonstrates tensor semantics only.

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No Tiny profitability result; explicit requirements before fund use

TINY PnL: NOT MEASURED

No Tiny-specific return target, execution policy, fill simulation, fee/slippage model, turnover, Sharpe, drawdown, or capacity estimate exists. Forecast skill cannot be translated into return percent.

Historical full-model diagnostic - not Tiny

Registered / post hoc trades	0 / 484
Post hoc gross / net	+2.16% / -68.07%
Gross edge / modeled round-trip cost	0.445 bp / 24.0 bp

Contextual warning only. It cannot be assigned to the Tiny checkpoint.

1 **DIRECT UTILITY**

Freeze executable return and spread-aware targets.

2 **STRONG BASELINES**

Freeze trees, quantile regression, and matched TCN.

3 **ROBUSTNESS**

Run 5 seeds and at least 5 purged rolling folds.

4 **EXECUTION**

Freeze bid/ask timing, fees, slippage, and impact.

5 **NEW EVIDENCE**

Freeze code, thresholds, and primary endpoints; then access 60+ new sessions.

PROVENANCE AND DECISION

`metrics 4d6023f80f78388bf93f... | protocol 9383fc46d76b8f02f131...`

`checkpoint 351ca13c779863b5f99c... | result 7ac87484ee6ff102acb8...`

RESEARCH CONTINUATION - NOT DEPLOYMENT